

$$s = cte \rightarrow M \rightarrow \text{if } M < 0.3 \rightarrow \text{incomp.}^1$$

Part II Compressible Flow

Objective:

The purpose of this part is to provide the student with the fundamental knowledge necessary to understand compressible aerodynamics.

$$0.3 \times 340 = 102 \text{ m/s}$$

So if $V_{\text{airplane}} > 100 \text{ m/s} \rightarrow \text{comp. flow}$

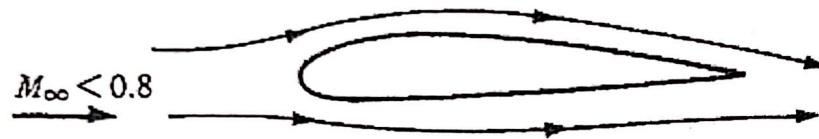
Flow Regimes

$$M = \text{Mach number} = \frac{\text{velocity magnitude}}{\text{speed of sound}} = \frac{|\vec{u}|}{a}$$

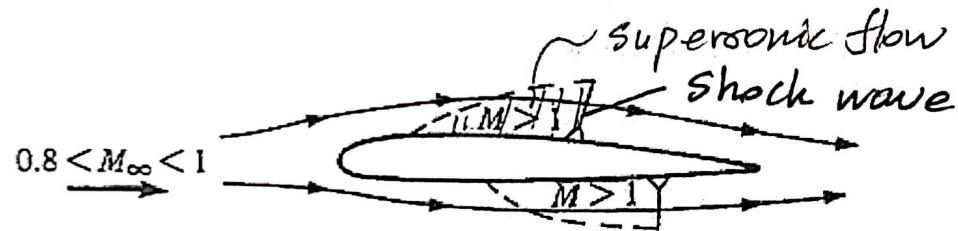
(Incompressible if $M < 0.3$)

$$a = \sqrt{\gamma R T}$$

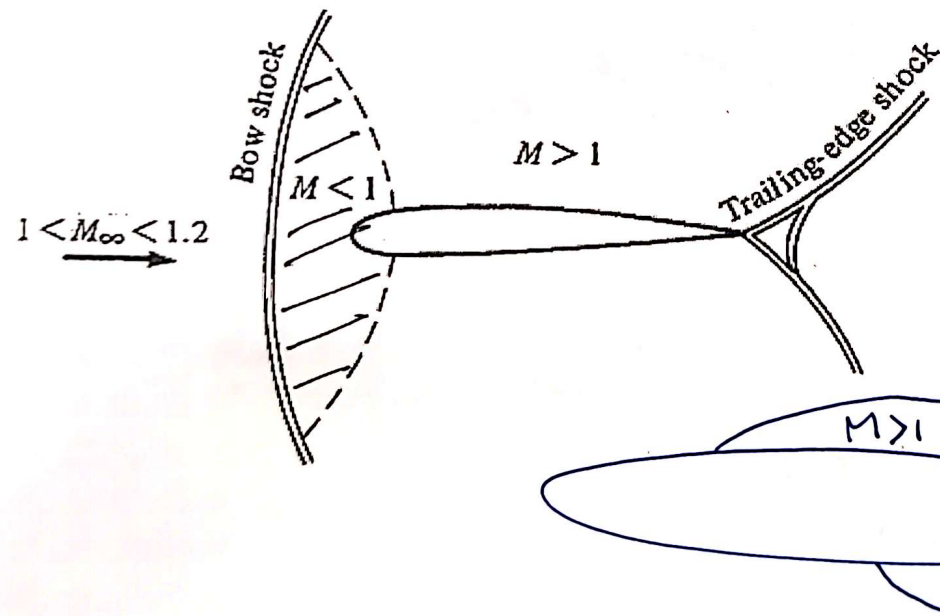
(to be proven later)



(a) Subsonic flow *every where*



(b) Transonic flow
with $M_\infty < 1$



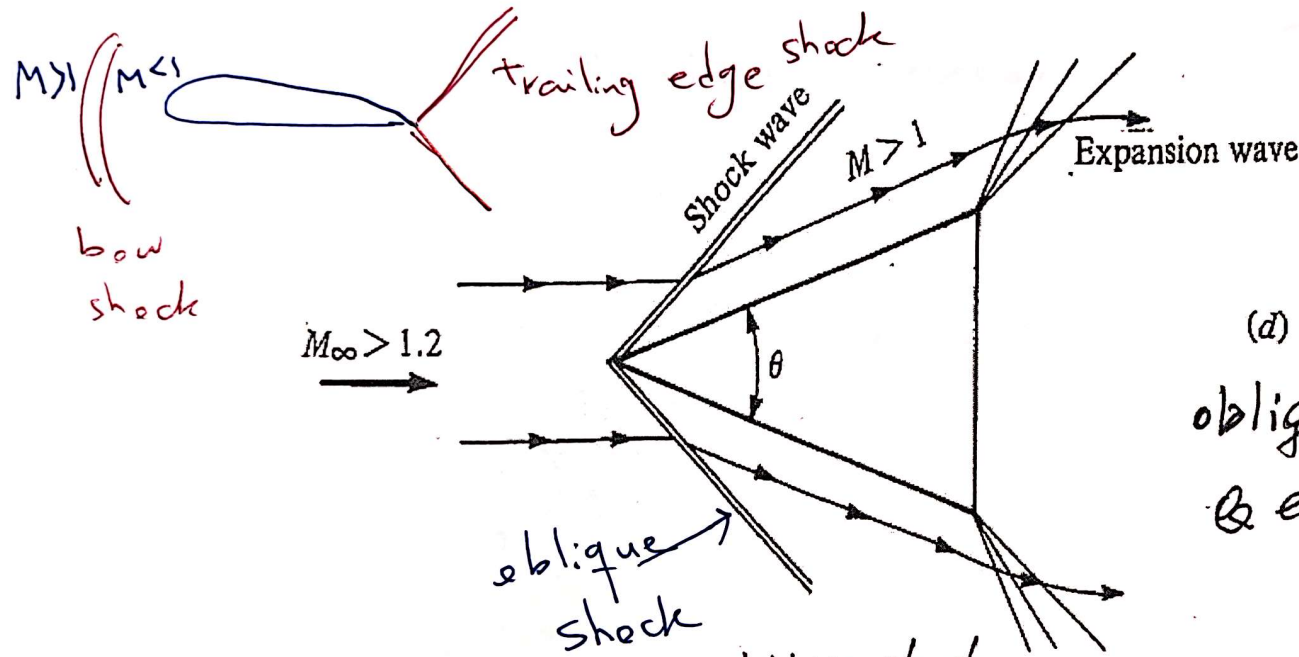
*regions of subsonic
& supersonic flow*

(c) Transonic flow
with $M_\infty > 1$

Normal shock wave with λ foot



3

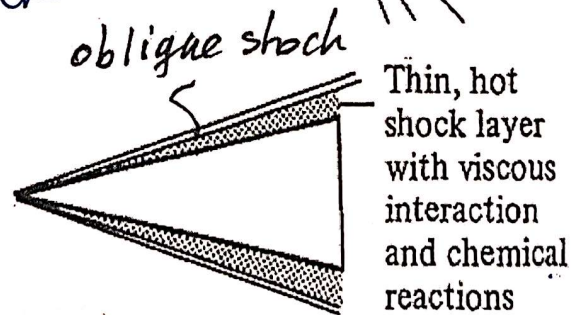


(d) Supersonic flow

oblique shocks

& expansion waves

$M_\infty > 5$



(e) Hypersonic flow

- high temperature near body
- possible chemical reactions dissociation
- discontinuous flow possible

(closer to the surface)